



Lecture 9

Functions

Announcements

- **HW 3** due Wednesday at 5pm
 - **Lab 4** due Friday
 - **Project 1** will be released on Friday
 - **HW 1** regrade requests are due Wednesday
 - My OH are also on [Zoom](#) today 4-6pm
-

Histograms

Area and Height

Area of bar = % in bin = Height x width of bin

- “How many individuals in the bin?” Use **area**.
- “How crowded is the bin?” Use **height**.

(Demo)

Charts Wrap Up

Summary

- **Line graph**: sequential data (over time, etc.)
- **Scatter plot**: relation between two numerical variables
- **Bar chart**: distribution of one categorical variable *or* relation between a categorical and a numerical variable
- **Histogram**: distribution of one numerical variable

(Demo)

Discussion Question

You have data about daily temperatures as shown. Which type of chart would show the answer to each question?

- Are there more cloudy than sunny days?
- What percentage of days have a high at least 72°?
- Do days with hotter highs tend to have hotter lows?

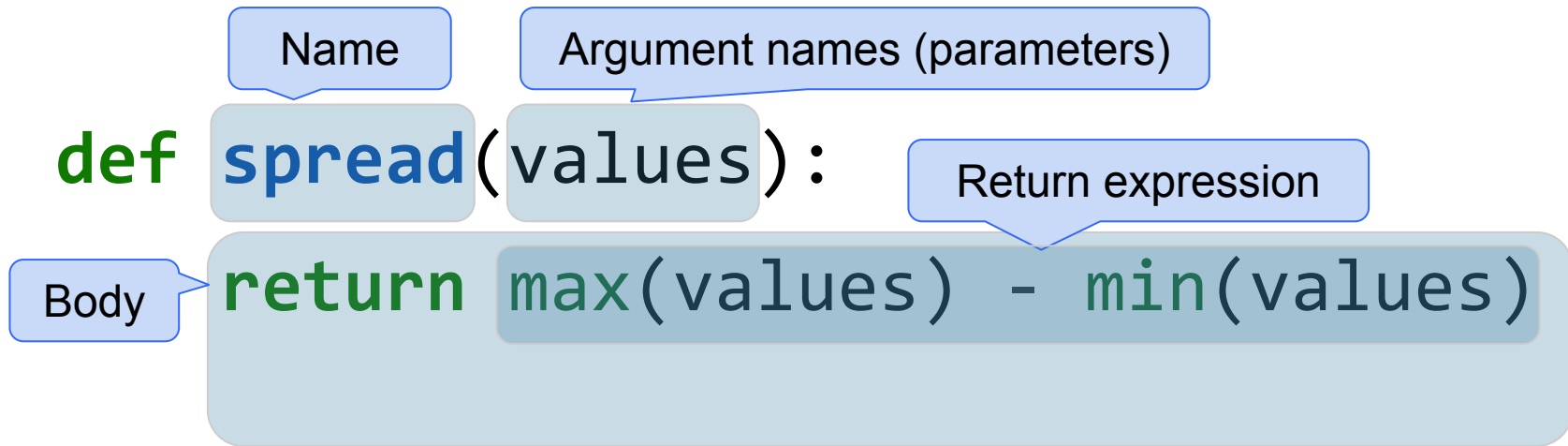
Day	High	Low	Sky condition
1	55.1	43.7	Cloudy
2	57.2	46	Sunny
3	56.8	45.9	Cloudy

... (362 rows omitted)

Defining Functions

Def Statements

User-defined functions give names to blocks of code



(Demo)

Discussion Question

What does this function do? What kind of input does it take? What output will it give? What's a reasonable name?

```
def f(s):  
    return np.round(s / sum(s) * 100, 2)
```

(Demo)

Apply

Apply

apply

1. Calls a function on every element in the input column(s)
2. Produces an array containing the output of the function on each input column element.
 - First argument: Function to apply
 - Other arguments: Specified input column(s)

```
table_name.apply(function_name, 'column_label(s)')
```

(Demo)
